

A stylized, high-contrast illustration of the Pacman character. The character is a large yellow circle with a wide, open black mouth and two large black eyes. It has orange ears and red shoes with white soles. The background is black.

Pacman Presentation

Recreating Pacman with Modern Web Technologies.

By

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Introduction / Project Overview.

Project Objective:

Recreate the classic Pacman game using JavaScript and HTML.

Deliver an interactive, engaging browser-based game that retains the charm of the original Pacman game.

Implementation:

- Utilized JavaScript for the game logic, animations, and user Interactions.
- Used HTML to structure the game interface, incorporating the canvas element to render visuals.

Project Scope:

- Develop Pacman's movement and controls.
- Implement Ghost AI with Dijkstra's algorithm
- Design an accurate game map that replicates the original
- Integrate sound effects to enhance the immersive experience.

Background and Motivation

History of Pacman:

Developed by Namco and released in 1980.

Became a cultural phenomenon due to its non-violent, maze-based gameplay.

Designed to appeal to a broad audience including women and families.

Significance in Gaming:

- Revolutionized the arcade game industry
- Known for its innovative gameplay mechanics and iconic characters.

Motivation:

- Broaden knowledge and practical experience in game development and web programming.
- Apply theoretical knowledge from the foundation year in a practical and engaging report.

Aims and Objectives

Primary Aim:

Develop a fully functional and engaging Pacman game that captures the essence of the original while incorporating modern programming techniques.

Specific Objectives:



Implement smooth and responsive controls for Pacman using keyboard input.



Develop unique behaviours for each ghost using Dijkstra's algorithm to enhance AI.



Design the game map to closely resemble the original Pacman maze.



Incorporate sound effects and animations to enhance the gameplay experience.

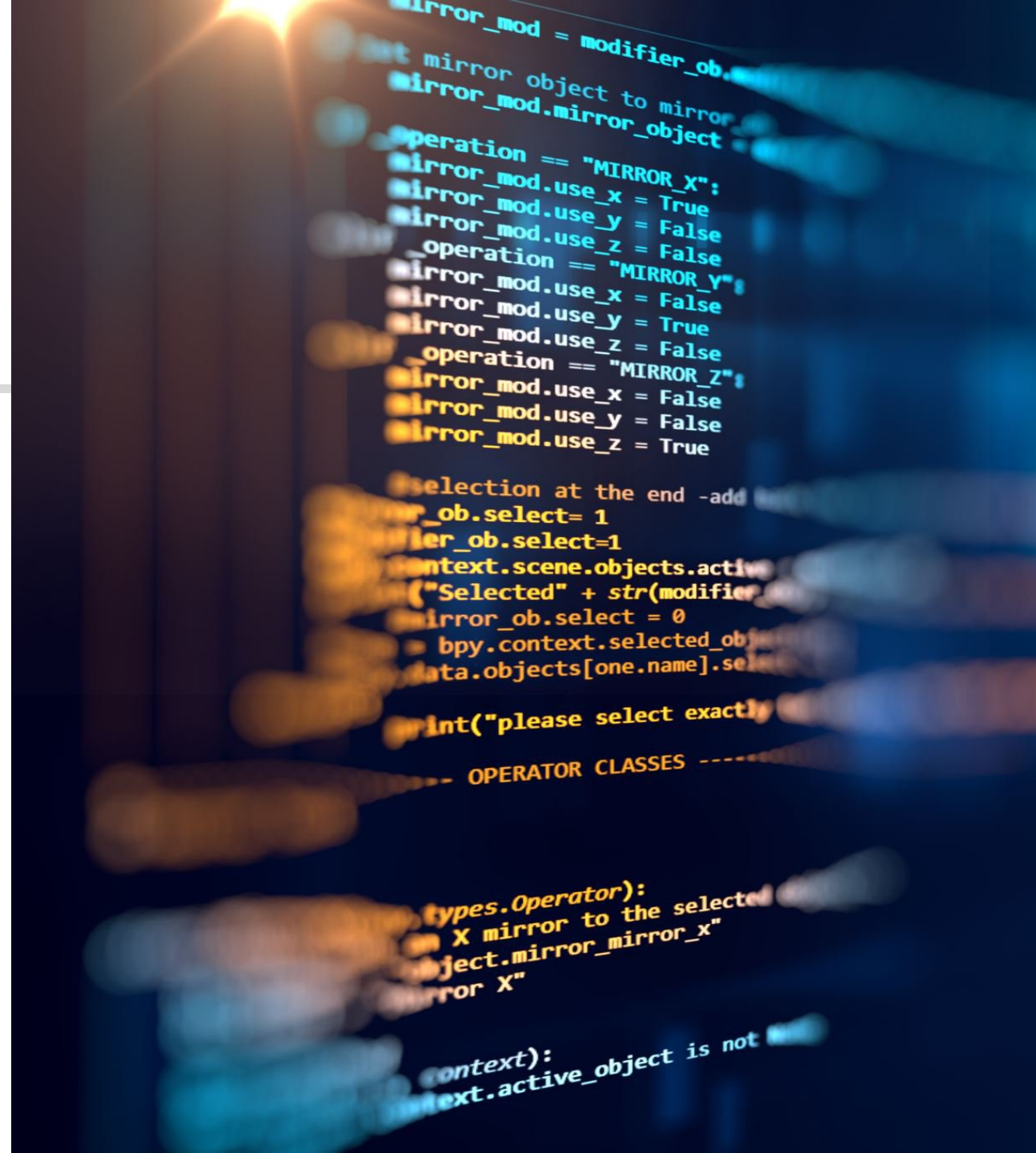
Game Development Frameworks

Technologies Used:

- HTML for structuring the game interface and integrating with other web technologies.
- JavaScript for handling game logic, animations, and user interactions.

Frameworks and Tools:

- Visual Studio Code for code editing and development.
- Chrome Developer Tools for debugging and testing.



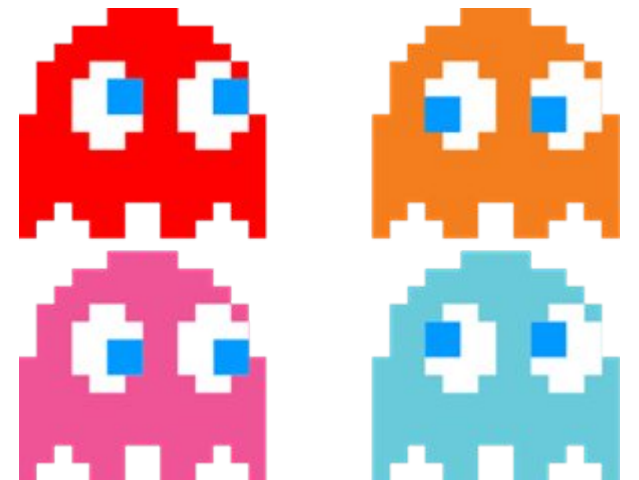
Sprite and Animation Techniques

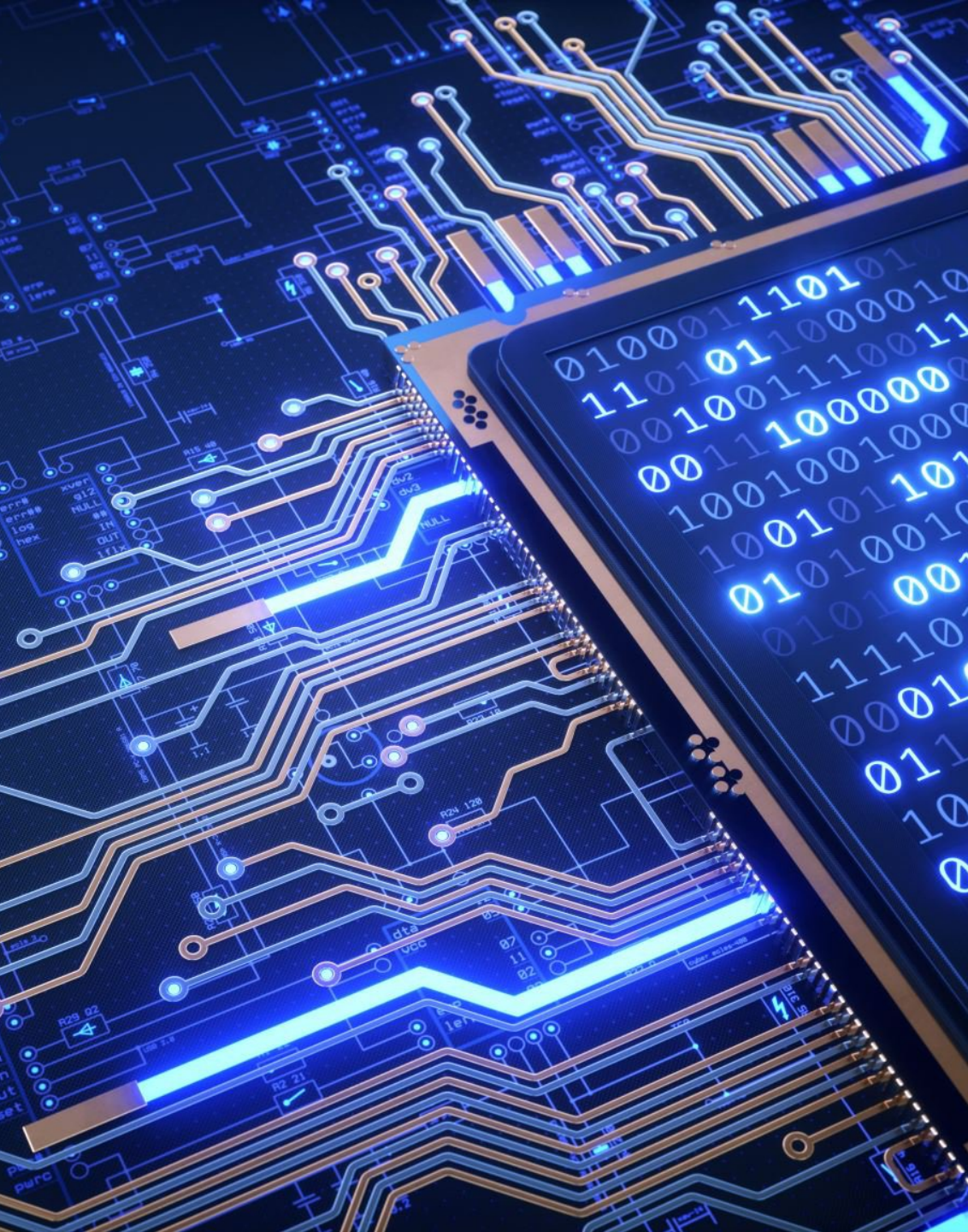
Sprites:

- Designed to replicate the classic look and feel of the original Pacman game.
- Created Individual frames for Pacman and the ghosts to stimulate movement.

Animation Techniques:

- Used a sequence of frames to animate Pacman's movement, similar to traditional cartoon animations.
- Applied similar techniques for ghost animations and state changes





Artificial Intelligence in Pacman

Ghost AI:

- Implemented using Dijkstra's algorithm for pathfinding within the maze.
- Ghosts follow specific patterns to mimic the original game's behaviour

AI behaviour:

- Each ghost has a unique behaviour pattern, creating a strategic and challenging gameplay experience.
- Used flowcharts to illustrate the decision-making process for ghost AI behaviour.

Design and Implementation

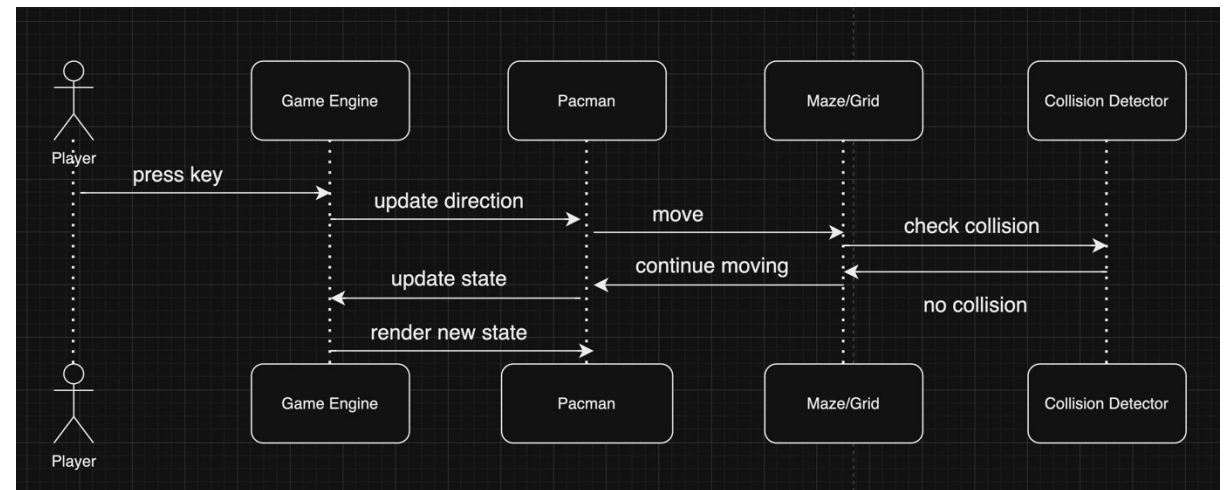
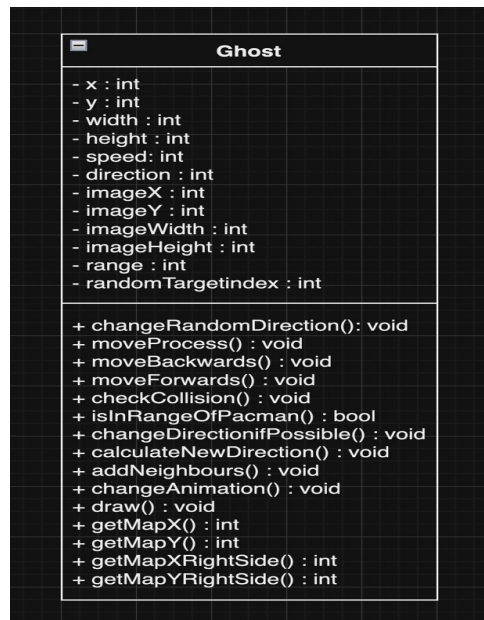
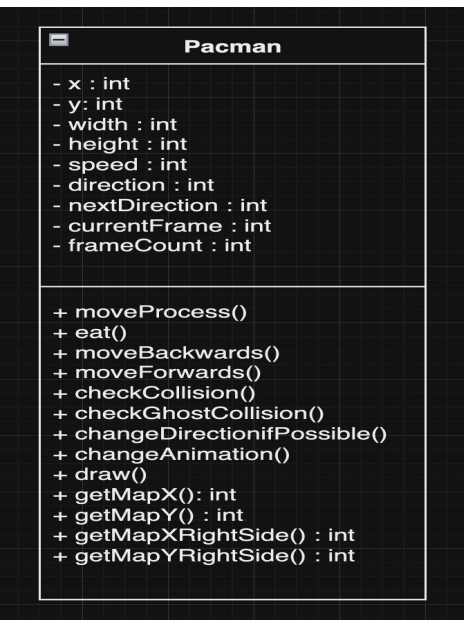
System Architecture:

- Adopted a molecular approach with distinct components such as the game engine, Pacman, ghosts, and user interface.

Ensured that each component interacts seamlessly to form a cohesive game.

Detailed Design:

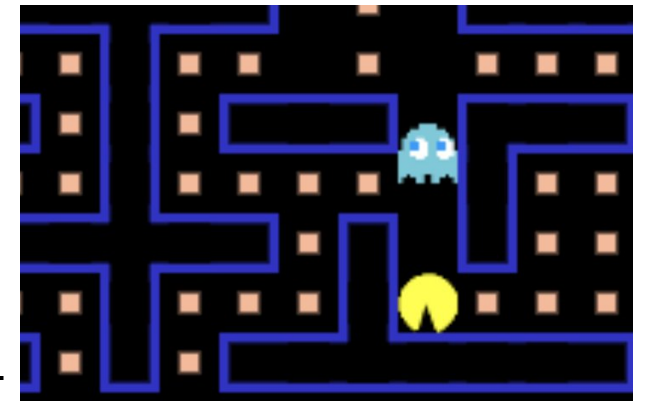
- Developed class diagrams for Pacman and Ghosts to represent their behaviours and interactions.
- Created sequence diagrams to show the flow of interactions between different game components.



Key Features

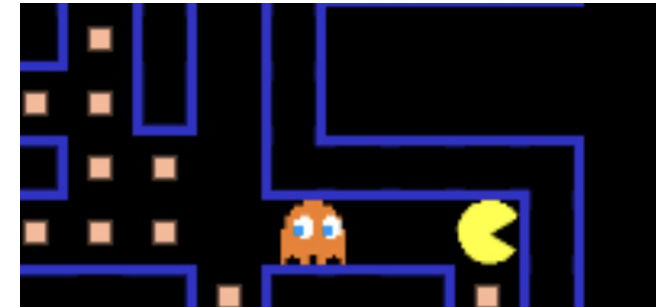
Pacman's Movement:

- Implemented smooth and responsive controls using keyboard input.
- Accurate collision detection with maze walls and pellets to maintain gameplay integrity.



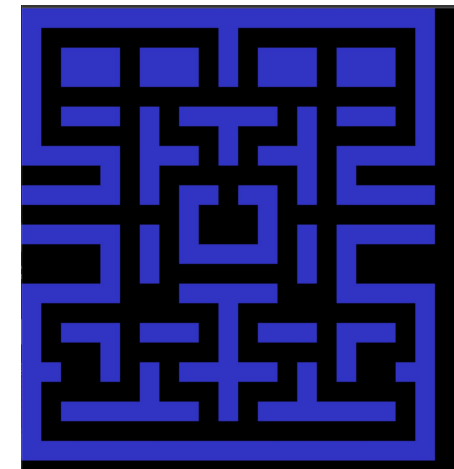
Ghost AI:

- Strategic movement to chase Pacman using Dijkstra's algorithm.
- Unique behaviour patterns for each ghost to mimic the original game.

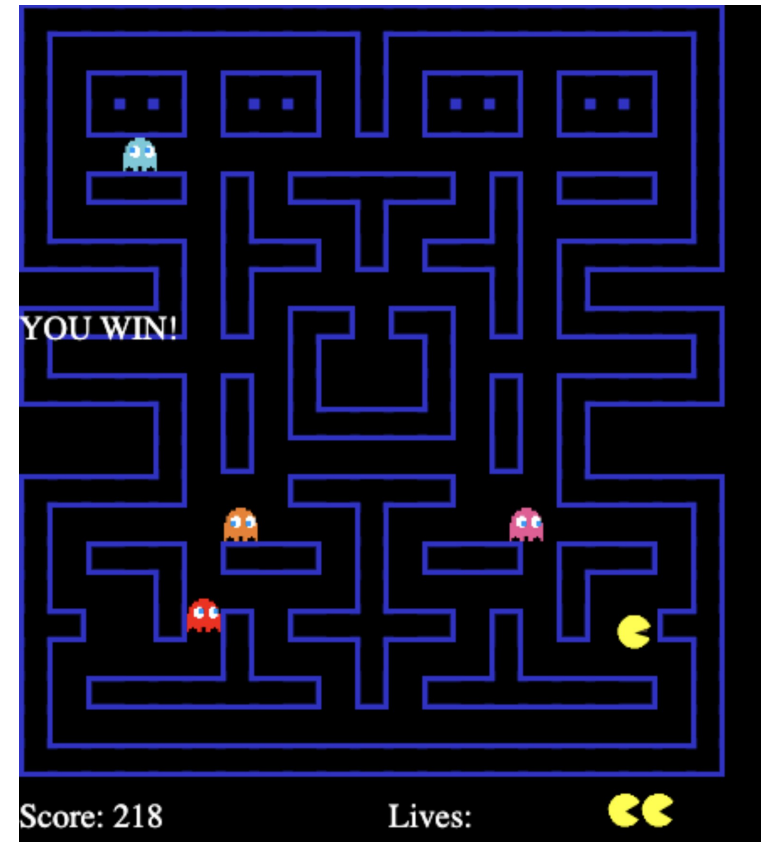
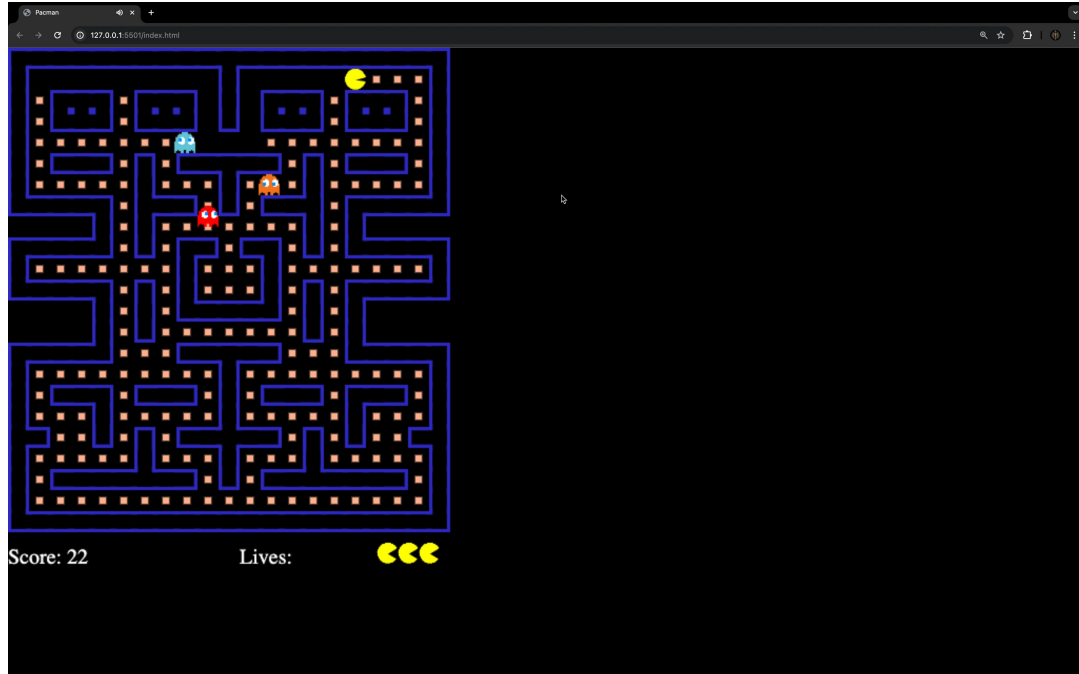


Game Environment:

- Designed the game map to closely resemble the original Pacman maze.
- Integrated sound effects to enhance the nostalgic feel of the game.



Live Demonstration



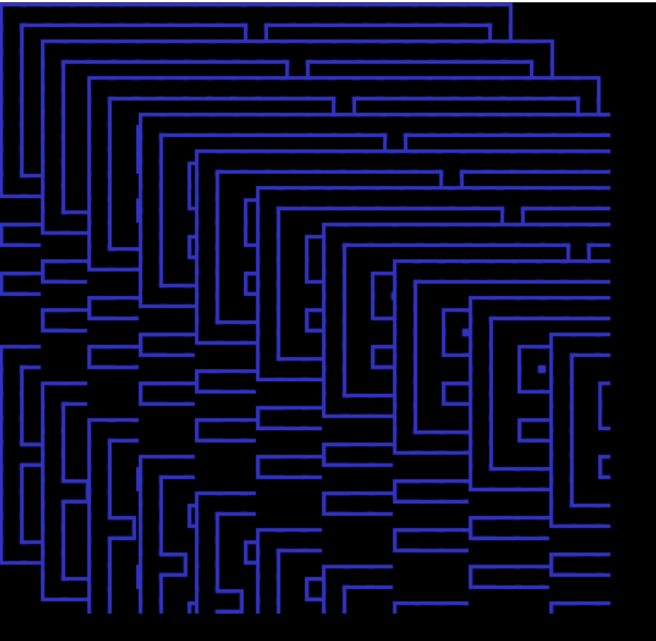
Challenges and Solutions

Power Pellets:

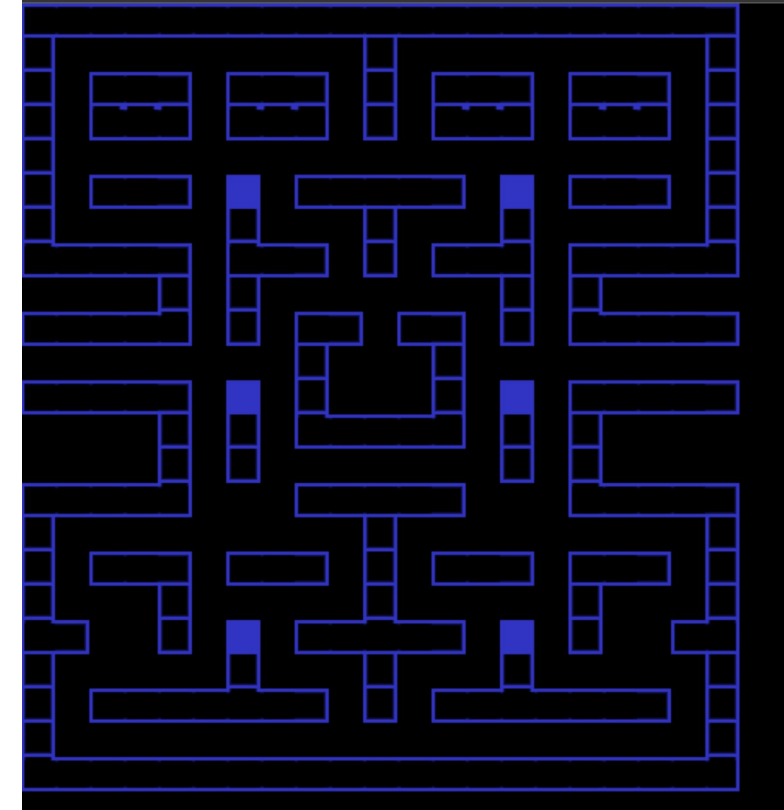
- Encountered issues with implementing the features to turn ghosts blue and make them vulnerable
- Frequent crashes and unexpected behaviours during development.

Solutions:

- Conducted extensive research to understand state management in JavaScript.
- Established a basic framework for future development of the power pellet feature.
- Improved error handling and debugging techniques to stabilize the game.



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✖ Uncaught TypeError: Failed to execute 'drawImage' on 'pacman.js:95  
'CanvasRenderingContext2D': The provided value is not of type  
'(CSSImageValue or HTMLCanvasElement or HTMLImageElement or  
HTMLVideoElement or ImageBitmap or OffscreenCanvas or  
SVGImageElement or VideoFrame)'.  
    at Pacman.draw (pacman.js:95:23)  
    at draw (game.js:62:12)  
    at gameLoop (game.js:50:5)  
    at game.js:131:1  
[Five Server] connecting... fiveserver.js:1  
[Five Server] connected. fiveserver.js:1  
153 Uncaught TypeError: Failed to execute 'drawImage' on 'pacman.js:95  
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Testing and Evaluation

Testing Plan:

Developed a comprehensive testing plan including unit, integration, and user acceptance testing.

Unit Testing :

- Verified functionality of individual components like Pacman's movement, ghost AI, and collision detection.

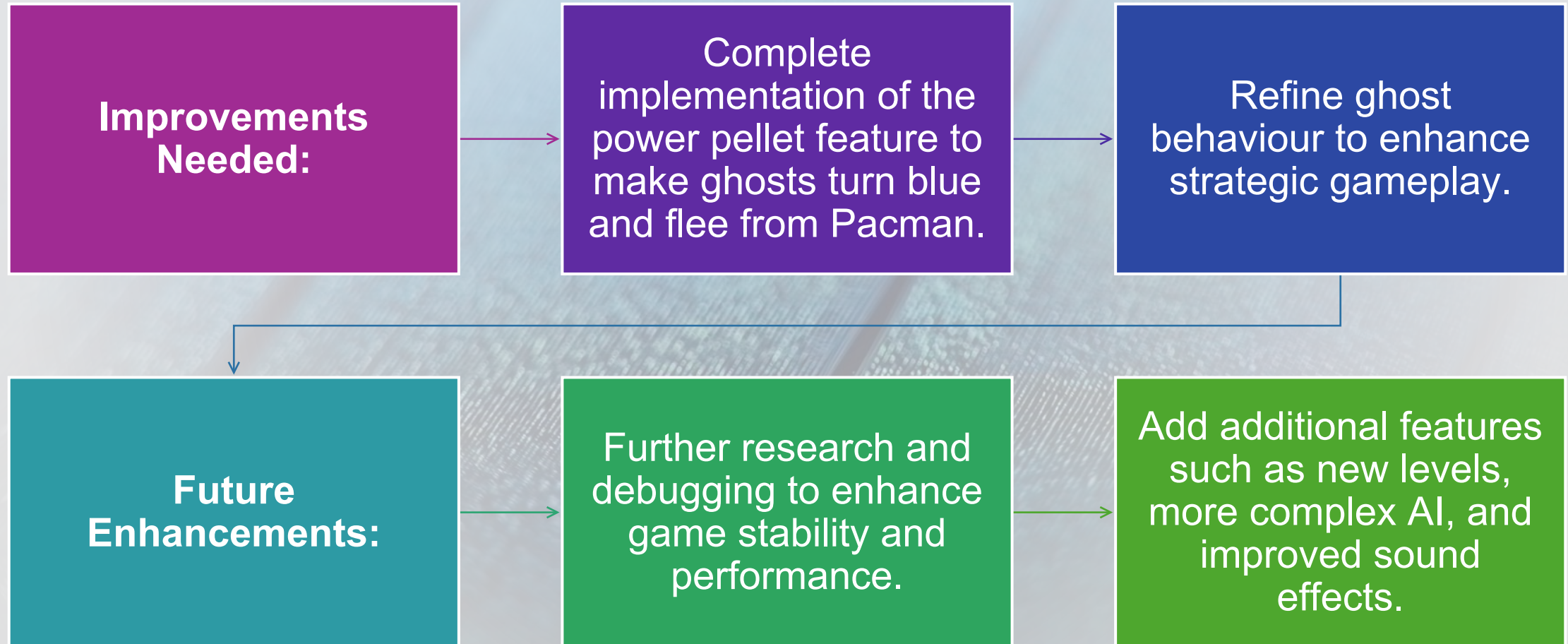
Integration Testing:

- Ensured that different components work together efficiently to form a cohesive game.

User Acceptance Testing:

- Collected feedback from users to validate the game's functionality and overall experience.

Future Work / Improvement



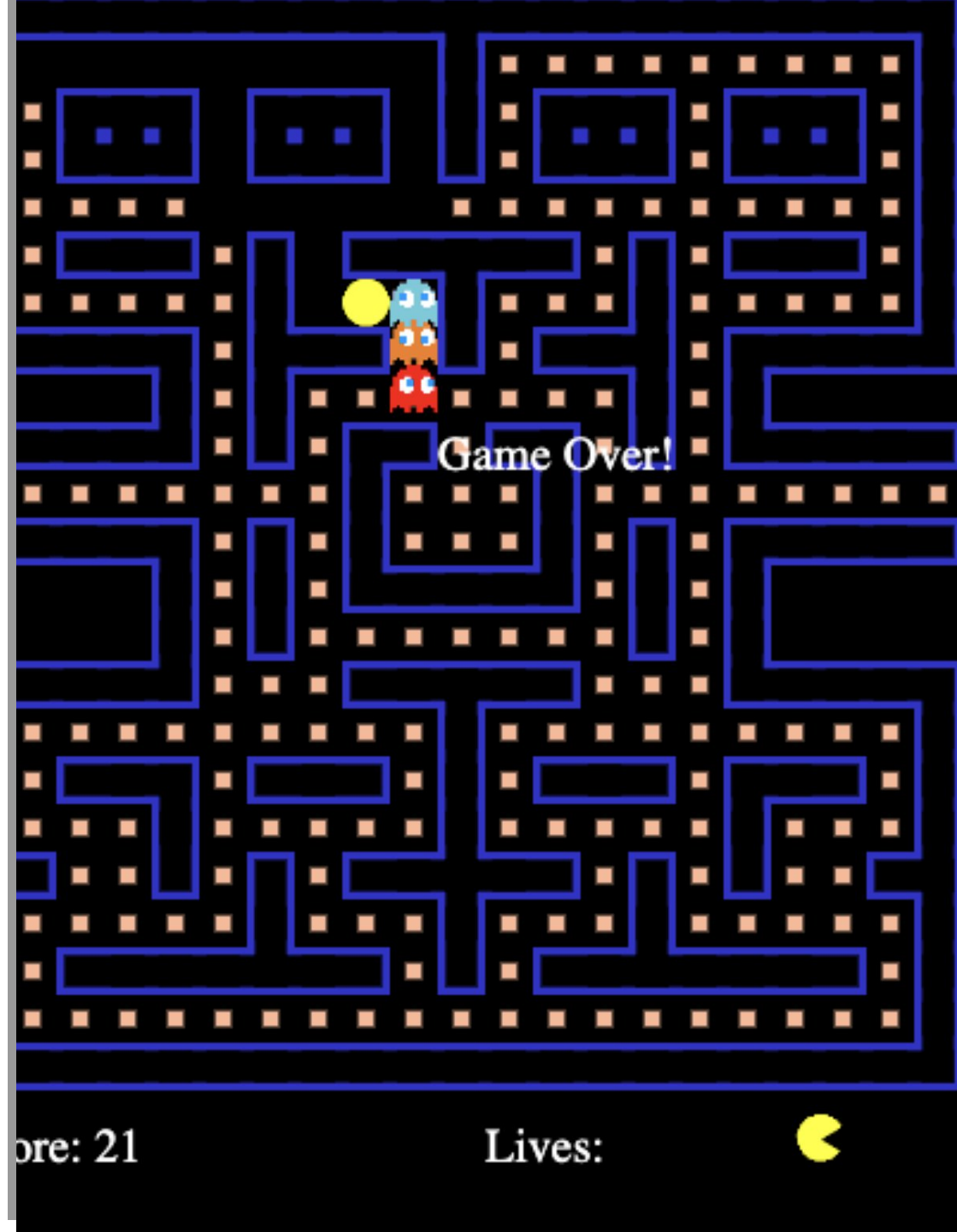
Conclusion

Summary:

Successfully developed a functional Pacman game using modern web technologies.

Demonstrated the potential of JavaScript and HTML in creating rich, interactive web-based games.

Showcased the application of foundational programming skills in a practical project.



Any
Questions?

